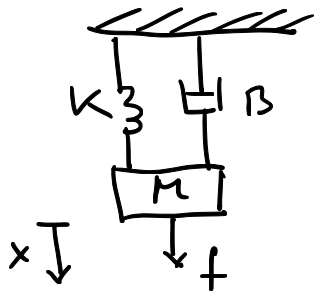
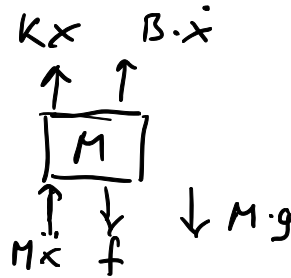


Exempel



Friläggning

→



Differenialekvationen enligt

D'Alembert lag:

$$-M\ddot{x} - Kx - B\dot{x} + f + M \cdot g = 0$$

$$\Rightarrow M\ddot{x} + B\dot{x} + Kx = f + M \cdot g \quad (1)$$

I viloläge: $\dot{x} = \ddot{x} = 0$, $f = 0$

$$\stackrel{(1)}{\Rightarrow} K \cdot x = M \cdot g$$

$$\Rightarrow x = \frac{M \cdot g}{K} \hat{=} x_0 = \frac{M \cdot g}{K}$$

Anta: $x = x_0 + z$

$$\Rightarrow \begin{cases} \dot{x} = \dot{z} \\ \ddot{x} = \ddot{z} \end{cases}$$

Substitution in i (1)

$$\stackrel{(1)}{\Rightarrow} M\ddot{z} + B\dot{z} + K(x_0 + z) = f + M \cdot g$$

$$\Rightarrow M\ddot{z} + B\dot{z} + Kz = -Kx_0 + M \cdot g + f$$

$$\Rightarrow M\ddot{z} + B\dot{z} + Kz = \underbrace{-K \frac{M \cdot g}{K} + M \cdot g + f}_{=0}$$

$$\Rightarrow M\ddot{z} + B\dot{z} + Kz = f$$

$$z = x - x_0!$$

